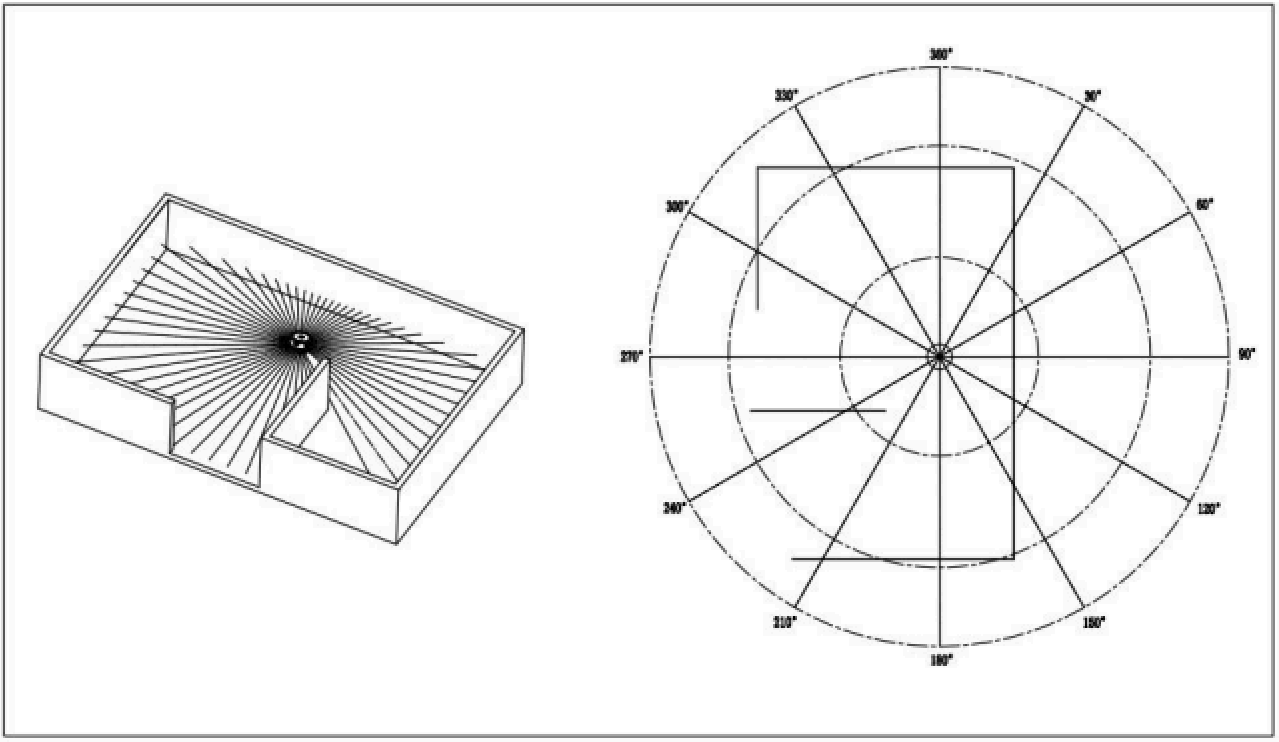
LD14

1.Principle and system introduction

LD14 is mainly composed of laser ranging core, wireless transmission unit, wireless communication unit, angle measurement unit and motor drive unit
Element and mechanical shell.
LD14 distance measuring core adopts triangulation technology, which can measure 2300 times per second. LD14 slave
An infrared laser is emitted from a fixed angle, and the laser is reflected to the receiving unit after encountering the target object. Through laser and target
The triangle relationship formed by the volume and the receiving unit can be used to calculate the distance.
After obtaining distance data, LD14 will fuse the angle values measured by the angle measurement unit to form point cloud data, and then
Wireless communication sends point cloud data to external interfaces. At the same time, the motor drive unit will drive the motor, which is closed loop controlled by PID algorithm
To the specified speed.

The environmental scanning diagram formed by LD14 point cloud data is intended as follows:



2.Specifications

2.1. Electrical and mechanical parameters

Parameter name	unit	Minimum value	Typical value	Maximum value	remarks
Input voltage	V	4.5V	5V	5.5V	
Starting current	mA	-	400	-	
Working current	mA	-	240	-	
Overall size	mm	96.3*59.8*38.8(length width height) tolerance ±0.3			
Machine weight	g	-	131	-	Connection cable not included
Communication interface	-	UART @ 115200			
UART high level	V	2.9	3.3	3.5	
The UART level is low	V	-0.3	0	0.4	
Driving motor	-	Dc brush motor			
Operating temperature	℃	-10	25	40	
Storage temperature	℃	-30	25	70	

2.2. Optical parameters

Parameter name	unit	Minimum value	Typical value	Maximum value	remarks
Laser wavelength	nm	775	793	800	Infrared band
Laser power	mW	-	10	-	
Laser safety grade	-	IEC-60825 Class 1			
pitch Angle	°	0	0.75	2	

2.3 performance parameter

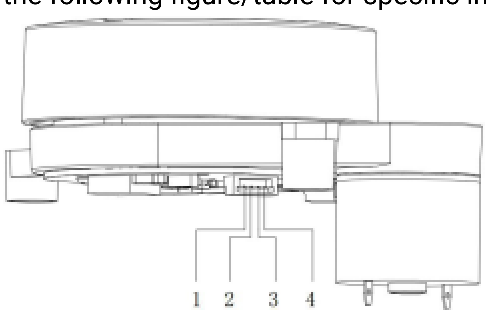
Parameter name	unit	Minimum value	Typical value	Maximum value	remarks
Ranging range	m	0.15	-	8	80% target reflectance
Sweep frequency	Hz	-	6	-	internal fixed speed
Ranging frequency	Hz	-	2300	-	fixed frequency
Ranging accuracy	mm	-	5	10	Ranging less than 1m
	-	-	1.5%	3%	ranging 1m-6m
			2%	5%	Ranging 6m to 8m
Angle error	°	-	1	2	
Angular resolution	°	-	1	-	
Ambient light resistance 〇注1	KLux	-	-	30	
Whole machine life	h	1500	-	-	

Note 1: The sun resistance performance is the data obtained in the internal test scenario. For details, please refer to the test scheme and test report of strong light interference.

3.data interface

3.1. Communications and Interfaces

LD14 uses a 1.25mm 4PIN connector to connect to the external system for power supply and data reception. See the following figure/table for specific interface definition and parameter requirements



serial number	Signal name	Type	escribe	least value	typical value	maximum
1	NC /PWM	Speed control	You can determine whether to control the speed externally	Refer to development manual	Refer to development manual	参Refer to development manual
2	GND	Negative	terminal of power supply	-	0V	-
3	Tx	Output	radar data output	0V	3.3V	3.5V
4	P5V	Positive	positive pole	4.5V	5V	5.5V

:

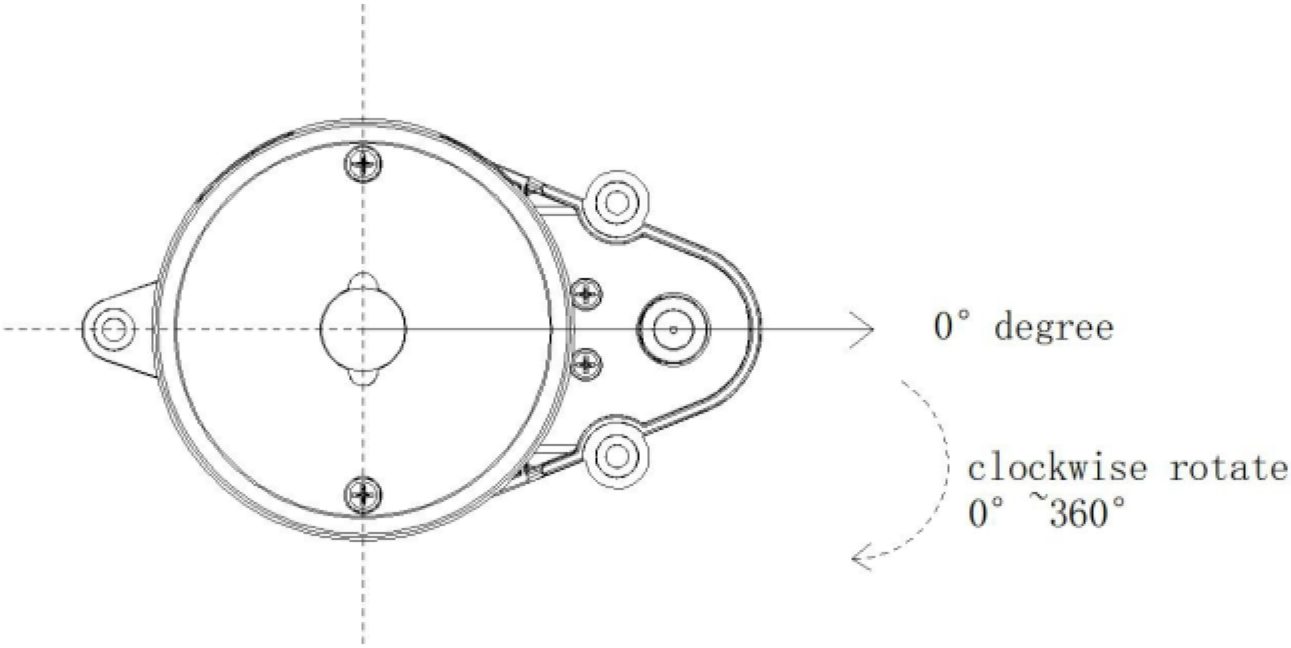
LD14 uses the standard asynchronous serial port (UART) for unidirectional transmission of data communication. The following table lists the transmission parameters

Baud rate	Data length	stop bit	parity check bit	flow control
115200	8 Bits	1	No	No

LD14 adopts unidirectional communication. After stable power-on, it starts to send measurement data without sending any instructions.

3.2. Definition of coordinate system

LD14 uses the left hand coordinate system, the center of rotation is the origin of coordinates, the line direction between the center of rotation and the center of the driving wheel is zero degree, and the rotation Angle increases along the clockwise direction. The details are shown in the following figure:



4.communication protocol

4.1 Packet form

LD14 adopts one-way communication. After stable operation, it starts to send measurement data packets without any instruction. The following figure shows the format of measurement packets.

Start character	VerLen	Radar speed		start Angle		data	end Angle		timestamp		CRC check
54H	1 Byte	LSB	MSB	LSB	MSB	......	LSB	MSB	LSB	MSB	1 Byte

initial identifier: length 1 Byte, fixed value of 0 x54 said the start of a packet;
VerLen: length 1 Byte, three said frame type, is fixed to 1, low five measuring points, said a package is fixed for 12, fixing the Byte value of 0 x2c;
radar speed: length 2 Byte, the unit for degrees per second;
starting Angle: the length of the 2 Byte, the unit of 0.01 degrees;
data: a measurement data length is 3 bytes, detailed analysis, please see the next section;
end Angle: the length of the Byte, 2 units of 0.01 degrees;
timestamp: length 2 Byte, the unit for ms, 30000, 30000 to count;
CRC check: The checksum of all previous data;

The data structure reference is as follows:

```
#define POINT_PER_PACK 12
#define HEADER 0x54
```

```
typedef struct   attribute ((packed)) { uint16_t distance;
uint8_t intensity;

} LidarPointStructDef;

typedef struct   attribute ((packed)) {
uint8_ t  header;
uint8_ t  ver_ len;
uint16_ t  speed;
uint16 t   start angle;
LidarPointStructDef point[POINT_ PER_ PACK];
uint16_ t   end_ angle;
uint16_ t   timestamp;
uint8_ t   crc8;
)LiDARFrame TypeDef;
```

CRC check calculation method is as follows:

```
static const uint8_t CrcTable[256]
={ 0x00, 0x4d, 0x9a, 0xd7, 0x79, 0x34,
0xe3,
0xae, 0xf2, 0xbf, 0x68, 0x25, 0x8b, 0xc6, 0x11, 0x5c, 0xa9, 0xe4, 0x33,
0x7e, 0xd0, 0x9d, 0x4a, 0x07, 0x5b, 0x16, 0xc1, 0x8c, 0x22, 0x6f, 0xb8,
0xf5, 0x1f, 0x52, 0x85, 0xc8, 0x66, 0x2b, 0xfc, 0xb1, 0xed, 0xa0, 0x77,
0x3a, 0x94, 0xd9, 0x0e, 0x43, 0xb6, 0xfb, 0x2c, 0x61, 0xcf, 0x82, 0x55,
0x18, 0x44, 0x09, 0xde, 0x93, 0x3d, 0x70, 0xa7, 0xea, 0x3e, 0x73, 0xa4,
0xe9, 0x47, 0x0a, 0xdd, 0x90, 0xcc, 0x81, 0x56, 0x1b, 0xb5, 0xf8, 0x2f,
0x62, 0x97, 0xda, 0x0d, 0x40, 0xee, 0xa3, 0x74, 0x39, 0x65, 0x28, 0xff,
0xb2, 0x1c, 0x51, 0x86, 0xcb, 0x21, 0x6c, 0xbb, 0xf6, 0x58, 0x15, 0xc2,
0x5f, 0xf1, 0xbc, 0x6b, 0x26, 0x7a, 0x37, 0xe0, 0xad, 0x03, 0x4e, 0x99,
0xd4, 0x7c, 0x31, 0xe6, 0xab, 0x05, 0x48, 0x9f, 0xd2, 0x8e, 0xc3, 0x14,
0x59, 0xf7, 0xba, 0x6d, 0x20, 0xd5, 0x98, 0x4f, 0x02, 0xac, 0xe1, 0x36,
0x7b, 0x27, 0x6a, 0xbd, 0xf0, 0x5e, 0x13, 0xc4, 0x89, 0x63, 0x2e, 0xf9,
0xb4, 0x1a, 0x57, 0x80, 0xcd, 0x91, 0xdc, 0x0b, 0x46, 0xe8, 0xa5, 0x72,
0x3f, 0xca, 0x87, 0x50, 0x1d, 0xb3, 0xfe, 0x29, 0x64, 0x38, 0x75, 0xa2,
0xef, 0x41, 0x0c, 0xdb, 0x96, 0x42, 0x0f, 0xd8, 0x95, 0x3b, 0x76, 0xa1,
0xec, 0xb0, 0xfd, 0x2a, 0x67, 0xc9, 0x84, 0x53, 0x1e, 0xeb, 0xa6, 0x71,
0x3c, 0x92, 0xdf, 0x08, 0x45, 0x19, 0x54, 0x83, 0xce, 0x60, 0x2d, 0xfa,
0xb7, 0x5d, 0x10, 0xc7, 0x8a, 0x24, 0x69, 0xbe, 0xf3, 0xaf, 0xe2, 0x35,
0x78, 0xd6, 0x9b, 0x4c, 0x01, 0xf4, 0xb9, 0x6e, 0x23, 0x8d, 0xc0, 0x17,
0x5a, 0x06, 0x4b, 0x9c, 0xd1, 0x7f, 0x32, 0xe5, 0xa8
};

uint8_t CalCRC8(uint8_t *p, uint8_t len){ uint8_t crc = 0;
uint16_t i;

```

```
for (i = 0; i < len; i++){

crc = CrcTable[(crc ^ *p++) & 0xff];

}

return crc;

}
```

4.2 Analysis of measurement data

Each measurement data point consists of a 2-byte distance value and a 1-byte confidence value, as shown in the figure below.

Start character	VerLen	Radar speed		start Angle		data	end Angle		timestamp		CRC check
54H	2CH	LSB	MSB	LSB	MSB	……	LSB	MSB	LSB	MSB	1Byte

Measuring point 1			Measuring point 2			…	Measuring point n		
Distance value		Signal strength	Distance value		signal strength		Distance value		signal strength
LSB	MSB	1 Byte	LSB	MSB	1 Byte	…	LSB	MSB	1 Byte

The unit of distance value is mm. The signal intensity value reflects the light reflection intensity, the higher the intensity, the larger the signal intensity value; The lower the intensity, the smaller the signal strength value.

The Angle value of each point is obtained by linear interpolation of the starting Angle and the ending Angle. The Angle calculation method is as follows:

```
step = (end_angle -- start_angle)/(len -- 1);
angle = start_angle + step*i;

len is the number of measurement points of a packet, and the value range of i is
```

4.3 Reference example

54 2C 68 08 AB 7E E0 00 E4 DC 00 E2 D9 00 E5 D5 00 E3 D3 00 E4 D0 00 E9 CD
00 E4 CA 00 E2 C7 00 E9 C5 00 E5 C2 00 E5 C0 00 E5 BE 82 3A 1A 50

We analyze it as follows:

Start character	VerLen	Radar speed		start Angle		data	end Angle		timestamp		CRC check
54H	2CH	68H	08H	ABH	7EH	……	BEH	82H	3AH	1AH	50H

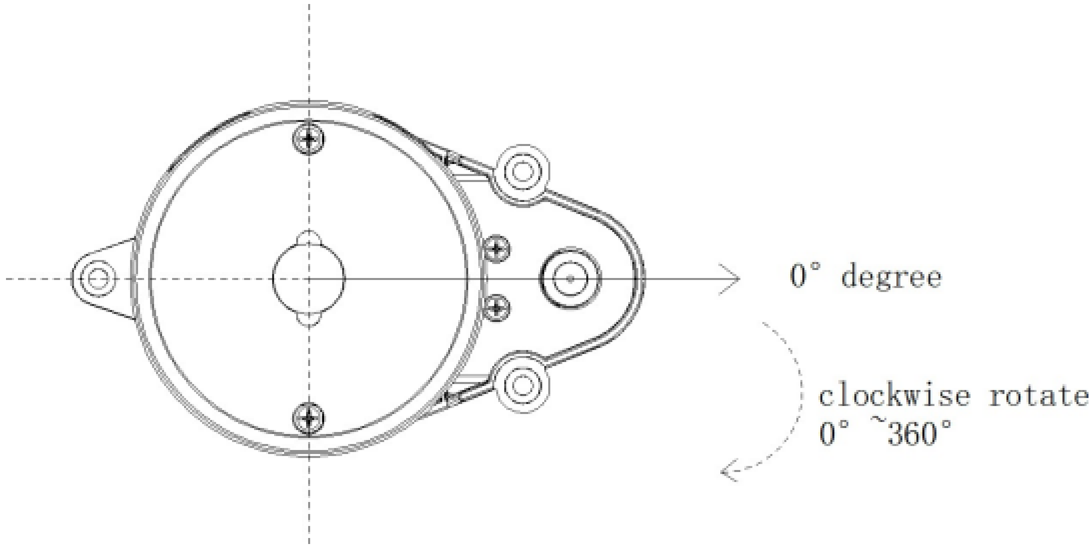
Measuring point 1			Measuring point 2			…	Measuring point 12		
Distance value		signal strength	Distance value		signal strength		Distance value		signal strength
E0H	00H	E4H	DCH	00H	E2H	…	B0H	00H	EAH

Field information	analysis
Radar speed	0868H＝2152 度/s;
Initial Angle	7EABH＝32427，即 324.27 度
End Angle	82BEH＝33470，即 334.7 度
Measuring point 1– Distance value	00E0H＝224mm
Measurement point 1– Signal strength	E4H＝228
Measuring point 2– distance value	00DCH＝200mm
Measurement point 2– Signal strength	00E2H＝226
---	…
Measuring point 12– distance value	00B0H＝176mm
Measurement point 12– Signal strength	EAH＝234

Due to the principle of trigonometric ranging, the above Angle value is uncorrected. See SDK for the detailed revision processSource transform.cpp in the Transform() function.

5.coordinate system

LD14 uses the left hand coordinate system, the center of rotation is the origin of coordinates, the right front of the sensor is defined as the zero degree direction, and the rotation Angle increases along the clockwise direction, as shown in the figure below.

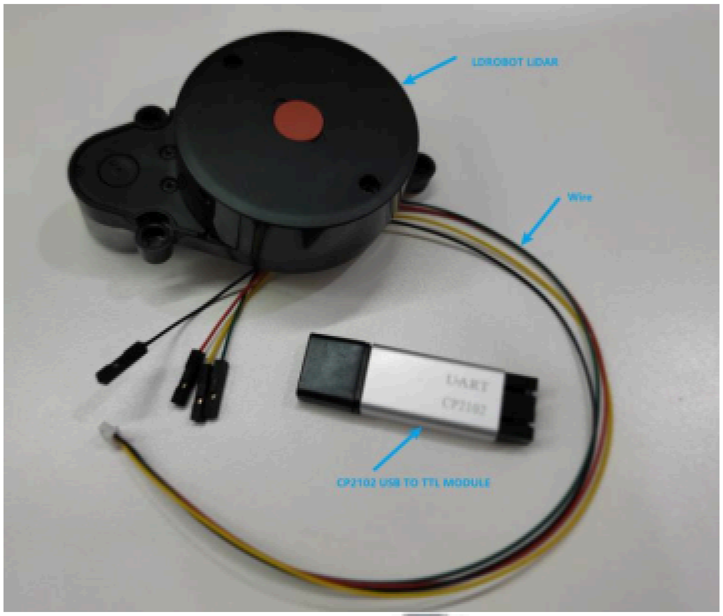


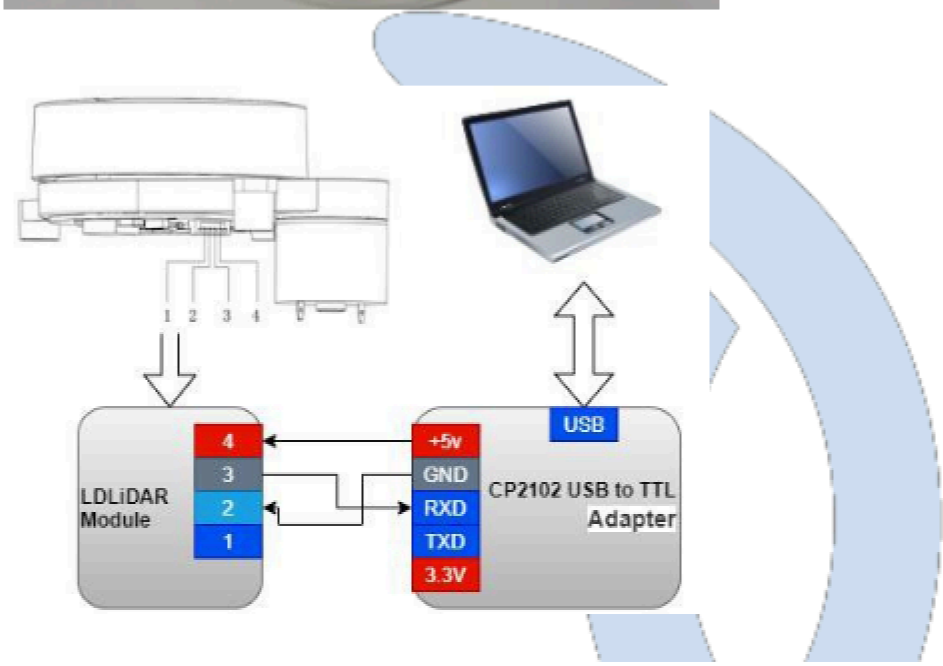
6.Develop the instructions for the suite

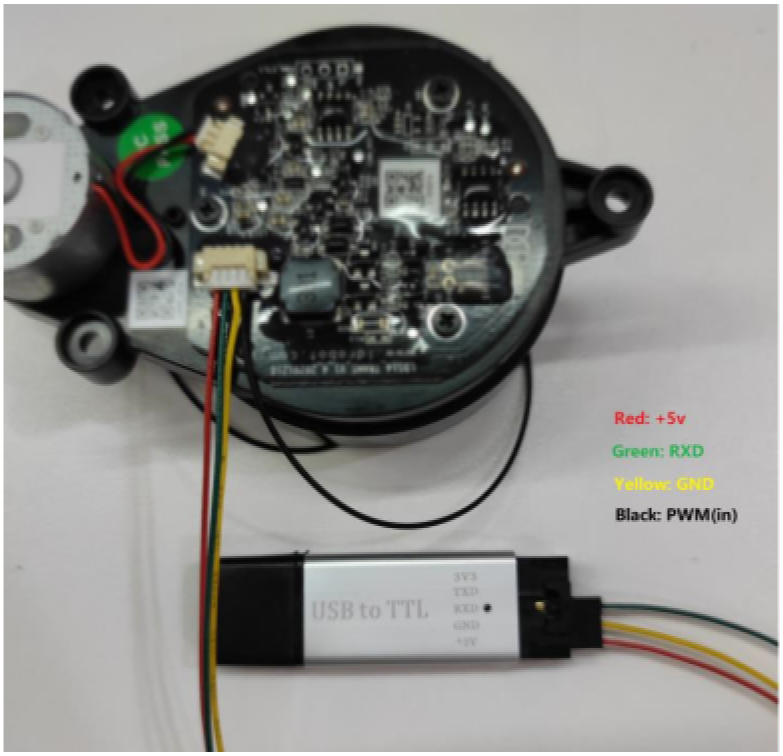
6.1. Evaluate the use of tools

6.1.1Hardware wire connection and description

1. Product, wire and USB adapter board, as shown in the figure below:







6.1.2. Installing the driver in windows

When evaluating our products under windows, it is necessary to install the serial port driver of USB adapter board, because CP2102 USB to serial port signal chip is adopted in the USB adapter board provided by our company. Its drivers can be downloaded from the official Silicon Labs website:

https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers

Or download it from our open source repository:

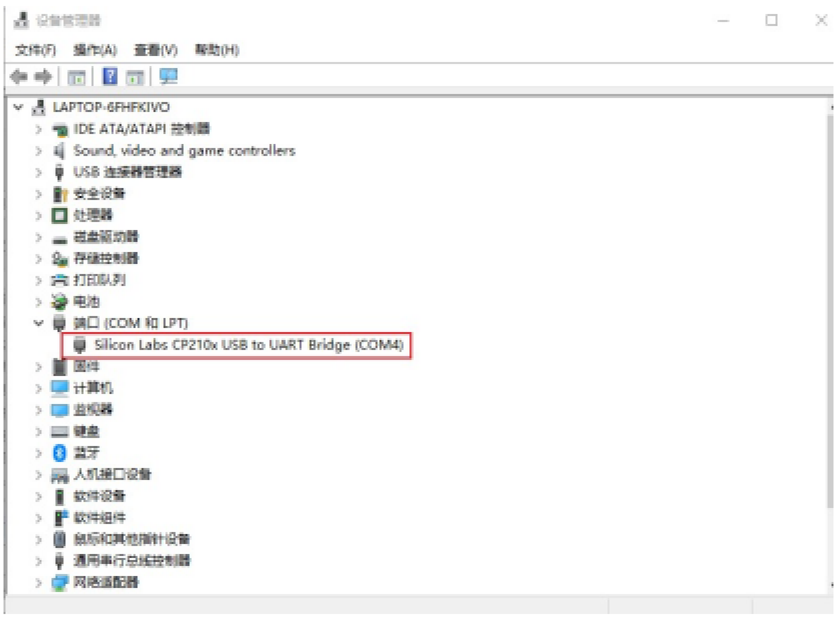
https://github.com/ldrobotSensorTeam/ld_desktop_tool/releases

Decompress the CP210x_Universal_Windows_Driver driver package, run the exe file in the driver installation package directory, and select X86(32-bit) or X64(64-bit) based on the Windows OS version.

名称	修改日期	类型	大小
arm	2018/12/8 0:06	文件夹	
arm64	2018/12/8 0:06	文件夹	
x64	2018/12/8 0:06	文件夹	
x86	2018/12/8 0:06	文件夹	
CP210x_Universal_Windows_Driver_ReleaseNotes...	2018/12/7 23:53	TXT 文件	20 KB
CP210xVCPInstaller_x64.exe	2018/5/8 6:05	应用程序	1,026 KB
CP210xVCPInstaller_x86.exe	2018/5/8 6:05	应用程序	903 KB
dpinst.xml	2018/5/8 5:46	XML 文件	12 KB
silabser.cat	2018/12/4 2:17	安全目录	13 KB
silabser.inf	2018/12/4 2:17	安装信息	10 KB
SLAB_License_Agreement_VCP_Windows.txt	2016/4/27 22:26	TXT 文件	9 KB

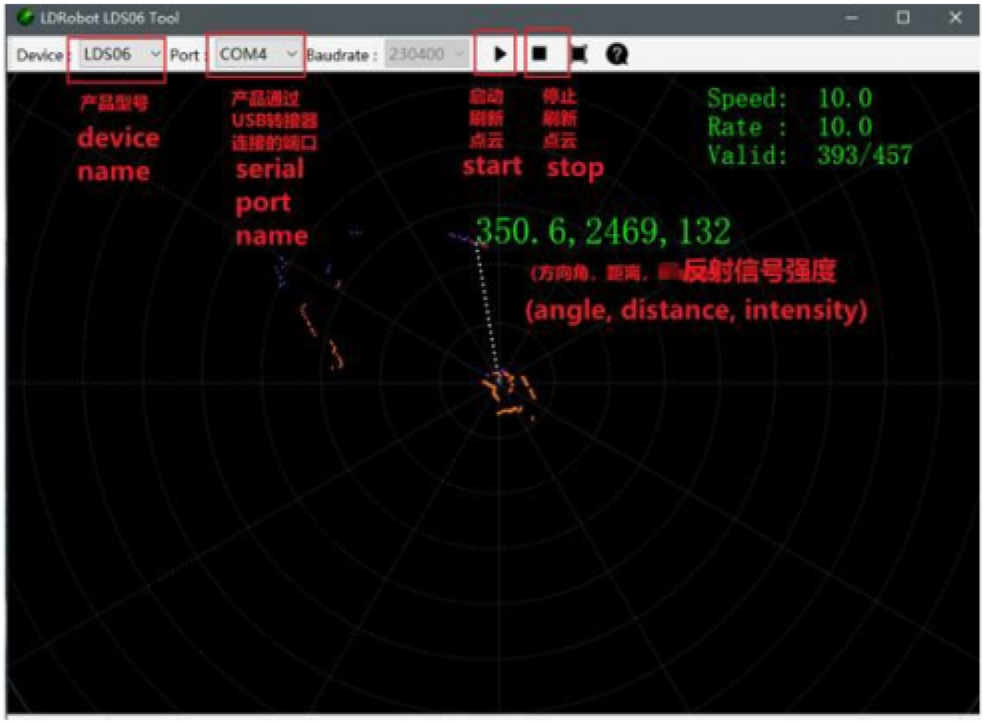
Double-click the exe file and follow the prompts to install it.

After the installation is complete, connect the USB adapter board in the development kit to the computer. You can right-click "My Computer" and choose "Properties". In the open "System" interface, select "Device Manager" from the left menu to enter the Device Manager and expand "Port". That is, the driver is installed successfully. The following figure shows COM4.



6.1.3. Evaluate the use of host computer software ld_desktop under windows

Our company provides point cloud visualization software ld_desktop for real-time scanning of this product. Developers can use this software to visually observe the scanning effect map of this product. Before using this software, ensure that the driver program of USB switching board of this product has been installed successfully, and connect this product with USB port of PC in Windows system. Then double-click LD_Deskdesktop.exe, select the corresponding product model and port number, and click the startup point cloud refresh button, as shown in the following figure.



In the figure above, Speed represents the laser radar scanning frequency (unit: Hz). Rate indicates the rate of lidar packet parsing; Valid Indicates the valid point of the lidar measurement circle.
The ld_desktop software binary package can be downloaded from our open source repository:
https://github.com/ldrobotSensorTeam/ld_desktop_tool/releases

6.2 Product 3D model file

The company provides 3D model files of this product in stp format to developers, which can be downloaded by visiting our open source warehouse:
https://github.com/ldrobotSensorTeam/Product_3D_model/releases

6.3. Ros-based operations in Linux

6.3.1.ROS Environment introduction and installation

ROS (Robot Operating System, "ROS" for short) is an open source meta-operating system for robots and middleware built on top of the Linux system. It provides the services expected of an operating system, including hardware abstraction, underlying device control, implementation of common functions, interprocess messaging, and package management. It also provides the tools and library functions needed to get, compile, write, and run code across computers. ROS each version of the installation steps, please refer to the ROS's official website:
http://wiki.ros.org/ROS/Installation

The ROS function pack of this product supports the following version environments:
ROS Kinetic(Ubuntu16.04);
ROS Melodic(Ubuntu18.04);
ROS Noetic(Ubuntu20.04).

6.3.2.Obtain the source code of the feature pack

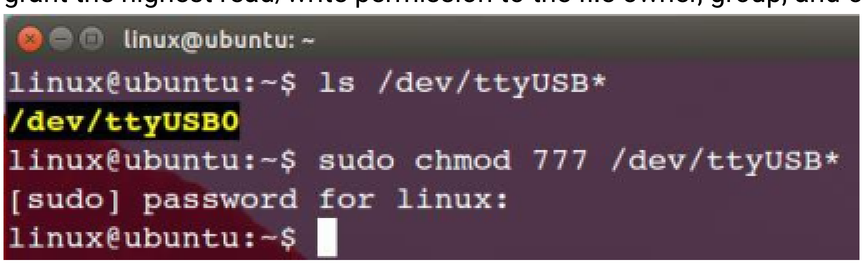
The source code of ROS function pack of this product is hosted on Github and Gitee repositories. You can download the source code of master or main branch by accessing the repository network link, or download it by git tool.

1. Warehouse website address
   https://gitee.com/ldrobotSensorTeam/ldlidar_sl_ros
   https://github.com/ldrobotSensorTeam/ldlidar_sl_ros
   git tool download operation

```
# First open the terminal interface, you can use the ctrl+alt+t shortcut
If you do not have git installed on your Ubuntu operating system, you can install
$ sudo apt-get install git
Download the ROS feature pack.
$ cd ~
$ mkdir -p ldlidar_ros_ws/src
$ cd ~/ldlidar_ros_ws/src
$ git clone https://gitee.com/ldrobotSensorTeam/ldlidar_sl_ros.git
# or
$ git clone https://github.com/ldrobotSensorTeam/ldlidar_sl_ros.git
```

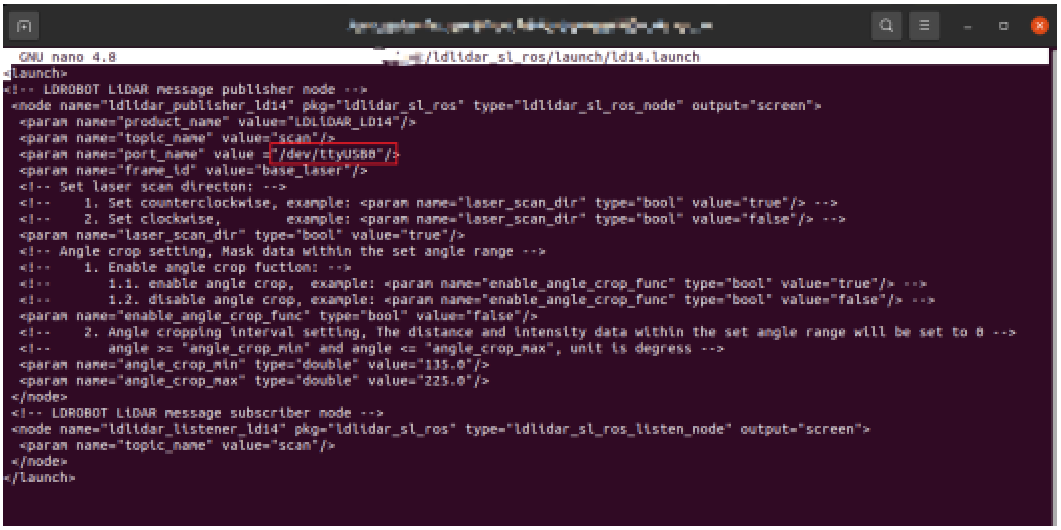
6.3.3. Setting device Permissions

First, the radar is connected to our switch module (CP2102 serial port switch module), and the module is connected to the computer. Then, open a terminal in ubuntu and type ls /dev/ttyUSB* to check whether the serial port device is connected. If a serial port device is detected, run the sudo chmod 777 /dev/ttyusb * command to grant the highest read/write permission to the file owner, group, and other users, as shown in the following figure.



Finally, change the port_name value in the ld14.launch file in the ~/ldldiar_ros_ws/src/ldlidar_sl_ros/launch/ directory. Take /dev/ttyusb0 as an example.

```
$ nano ~/ldlidar_ros_ws/src/ldldiar_sl_ros/launch/ld14.launch
```



```
GNU nano 4.8                    /ldlidar_sl_ros/launch/ld14.launch
<launch>
<!-- LDROBOT LiDAR message publisher node -->
 <node name="ldlidar_publisher_ld14" pkg="ldlidar_sl_ros" type="ldlidar_sl_ros_node" output="screen">
  <param name="product_name" value="LDLiDAR_LD14"/>
  <param name="topic_name" value="scan"/>
  <param name="port_name" value ="/dev/ttyUSB0"/>
  <param name="frame_id" value="base_laser"/>
  <!-- Set laser scan directon: -->
  <!--    1. Set counterclockwise, example: <param name="laser_scan_dir" type="bool" value="true"/> -->
  <!--    2. Set clockwise,        example: <param name="laser_scan_dir" type="bool" value="false"/> -->
  <param name="laser_scan_dir" type="bool" value="true"/>
  <!-- Angle crop setting, Mask data within the set angle range -->
  <!--    1. Enable angle crop fuction: -->
  <!--       1.1. enable angle crop,  example: <param name="enable_angle_crop_func" type="bool" value="true"/> -->
  <!--       1.2. disable angle crop, example: <param name="enable_angle_crop_func" type="bool" value="false"/> -->
  <param name="enable_angle_crop_func" type="bool" value="false"/>
  <!--    2. Angle cropping interval setting, The distance and intensity data within the set angle range will be set to 0 -->
  <!--       angle >= "angle_crop_min" and angle <= "angle_crop_max", unit is degress -->
  <param name="angle_crop_min" type="double" value="135.0"/>
  <param name="angle_crop_max" type="double" value="225.0"/>
 </node>
 <!-- LDROBOT LiDAR message subscriber node -->
 <node name="ldlidar_listener_ld14" pkg="ldlidar_sl_ros" type="ldlidar_sl_ros_listen_node" output="screen">
  <param name="topic_name" value="scan"/>
 </node>
</launch>
```

Linux nano editor: Ctrl + O to save the edit file; Ctrl + X to exit the editing screen.

6.3.4.Compilation and environment setup

1. Compile and build product function packs with catkin compilation system:

```
$ cd ~/ldlidar_ros_ws
$ catkin_make
```

2. Function pack environment variable setting:

After the compilation is complete, you need to add the related files generated by the compilation to the environment variables so that the ROS environment can be identified. Run the following command. The command is used to temporarily add environment variables to the terminal, which means that you need to run the following command again if you restart a new terminal.

```
$ cd ~/ldlidar_ros_ws
$ source devel/setup.bash
```

To permanently remove the need to run the preceding command to add environment variables after restarting the terminal, perform the following operations.

```
$ echo “source ~/ldlidar_ros_ws/devel/setup.bash” >> ~/.bashrc
$ source ~/.bashrc
```

6.3.5. Run the program

Use roslaunch tool to start the radar node and execute the following command.

```
$ roslaunch ldlidar_sl_ros ld14.launch
```

6.3.6.Rviz displays radar point cloud

Rviz is a 3D visualization tool within the ROS framework that allows radar data to be displayed. in

After roslaunch successfully runs the radar node, open a new terminal (Ctrl+Alt+T) and enter the following command:

```
$ rosrun rviz rviz
```

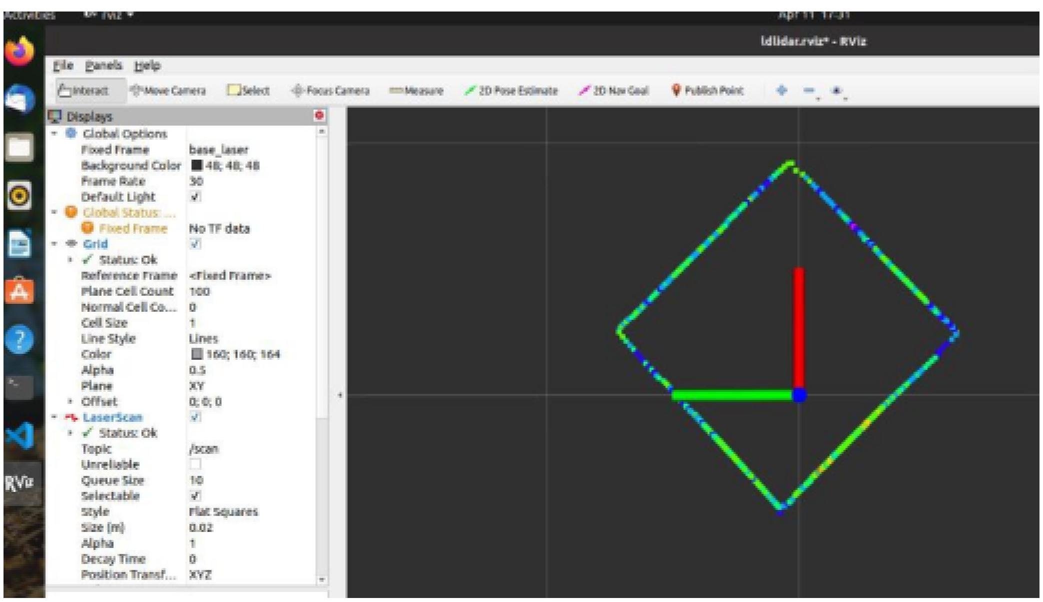
After the interface is started, click file->open->Config on the menu bar of the interface, and then select:

~/ldlidar_ros_ws/src/ldlidar_sl_ros/rviz/ directory ldlidar.rviz to open the configuration file

After ldlidar.rviz, click on LaserScan Topic and select /scan.

Since the point cloud data of our radar products follow the left-handed coordinate system, while the radar point cloud data displayed by Rviz follow the right-handed coordinate system, we carried out the coordinate system transformation of the product point cloud data in the function package source toLaserScan function, so that it can normally display the point cloud in Rviz.

In addition, there is a certain compatibility problem with the Rviz configuration file between different ROS versions. If you open the ldlidar.rviz file and set relevant parameters, you still cannot view the lidar point cloud or the Rviz tool crashes. You can solve this problem by creating another Rviz configuration file that mimics the Settings of ldlidar.rviz.



6.4. Ros2-based operations in Linux

6.4.1.ROS2 Environment Introduction and Installation

ROS (Robot Operating System, "ROS" for short) is an open source meta-operating system for robots and middleware built on top of the Linux system. It provides the services expected of an operating system, including hardware abstraction, underlying device control, implementation of common functions, interprocess messaging, and package management. It also provides the tools and library functions needed to get, compile, write, and run code across computers. Since the

Much has changed in robotics and the ROS community since ROS was launched in 2007. The ROS2 project aims to adapt to these changes, taking advantage of the strengths of ROS1 and improving on the weaknesses. For details about how to install ROS2, see

ROS2 official website: https://docs.ros.org/en/foxy/Installation.html

The ROS2 function package of this product can be used in the ROS2 Foxy(Ubuntu20.04) environment.

6.4.2. Obtain the source code of the feature pack

The source code of ROS2 function pack of this product is hosted on the repositories of Github and Gitee. You can download the source code of master or main branch by accessing the network link of the repositories, or download it by git tool.

1. Warehouse website address

 https://gitee.com/ldrobotSensorTeam/ldlidar_sl_ros2

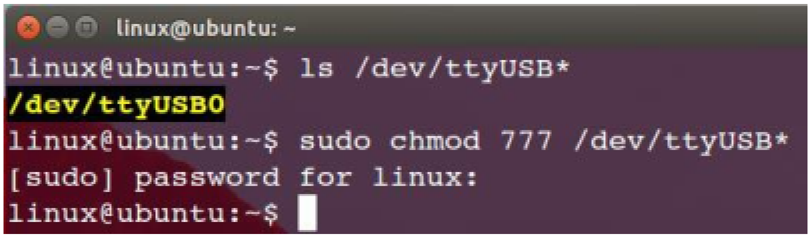
 https://github.com/ldrobotSensorTeam/ldlidar_sl_ros2

2)git tool download operation

```
# First open the terminal interface, you can use the ctrl+alt+t shortcut
If you do not have git installed on your Ubuntu operating system, you can install
$ sudo apt-get install git
# ROS2
$ cd ~
$ mkdir -p ldlidar_ros2_ws/src
$ cd ~/ldlidar_ros2_ws/src
$ git clone https://gitee.com/ldrobotSensorTeam/ldlidar_sl_ros2.git
# or
$ git clone https://github.com/ldrobotSensorTeam/ldlidar_sl_ros2.git
```

6.4.3. Setting device Permissions

First, the radar is connected to our switch module (CP2102 serial port switch module), and the module is connected to the computer. Then, open a terminal in ubuntu and type ls /dev/ttyUSB* to check whether the serial port device is connected. If a serial port device is detected, run the sudo chmod 777 /dev/ttyusb * command to grant the highest read/write permission to the file owner, group, and other users, as shown in the following figure.



Finally, change the port_name value in the ld14.launch.py file in the ~/ldldiar_ros2_ws/src/ldlidar_sl_ros2/launch/ directory. Take /dev/ttyusb0 as an example.

```
$ nano ~/ldlidar_ros2_ws/src/ldldiar_sl_ros2/launch/ld14.launch.py
```

Linux nano editor: Ctrl + O to save the edit file; Ctrl + X to exit the editing screen.

6.4.4 Compilation and environment setup

1.Compile and build product feature packs using colcon compilation system

```
$ cd ~/ldlidar_ros2_ws
$ colcon build
```

2.Feature pack environment variable Settings:
After the compilation is complete, related files generated by compilation need to be added to environment variables so that ROS2 environment can be identified. Run the following command, which is to temporarily add environment variables to the terminal, which means that if you reopen a new terminal, you need to run the following command again.

```
$ cd ~/ldlidar_ros2_ws
$ source install/setup.bash
```

To permanently remove the need to run the preceding command to add environment variables after restarting the terminal, perform the following operations.

```
$ echo “source ~/ldlidar_ros2_ws/install/setup.bash” >> ~/.bashrc
$ source ~/.bashrc
```

6.4.5. Run the program

Use ros2 launch tool to start the radar node and execute the following command.

```
$ ros2 launch ldlidar_sl_ros2 ld14.launch.py
```

6.4.6.Rviz2 Displays radar point cloud

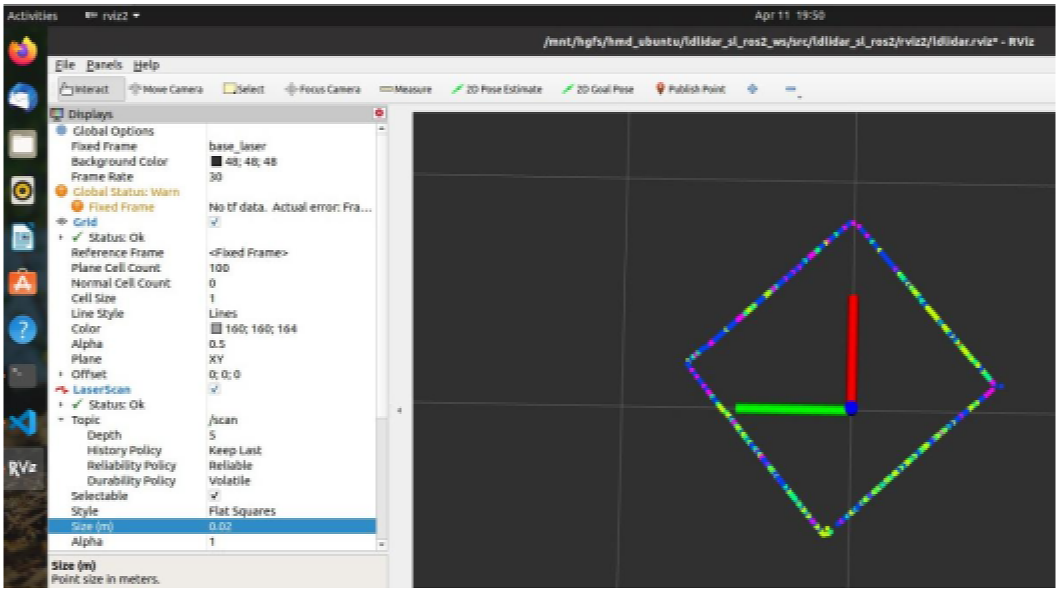
Rviz2 is a 3D visualization tool within the ROS2 framework where radar data can be displayed. in
After ros2 launch successfully runs the radar node, open a new terminal (Ctrl+Alt+T) and enter the following command:

```
$ rviz2
```

After the interface is started, click file->open->Config on the menu bar of the interface, and then select:

~/ldlidar_ros2_ws/src/ldlidar_sl_ros2/rviz2/ directory ldlidar.rviz file, open the configuration file ldlidar.rviz, click LaserScan Topic and select /scan.
Since the point cloud data of our radar products follow the left-handed coordinate system, while the radar point cloud data displayed by Rviz2 follow the right-handed coordinate system, we carried out coordinate transformation of the product point cloud data in the function package source toLaserScan function, so that it can

normally display point cloud in Rviz2.



6.5. SDK Usage in Linux

6.5.1. Obtain the SDK source code

The Linux SDK source code of this product is hosted on Github and Gitee warehouses. You can download the source code of master or main branch by accessing the warehouse network link, or download it by git tool.

1. Warehouse website address
   https://gitee.com/ldrobotSensorTeam/ldlidar_sl_sdk

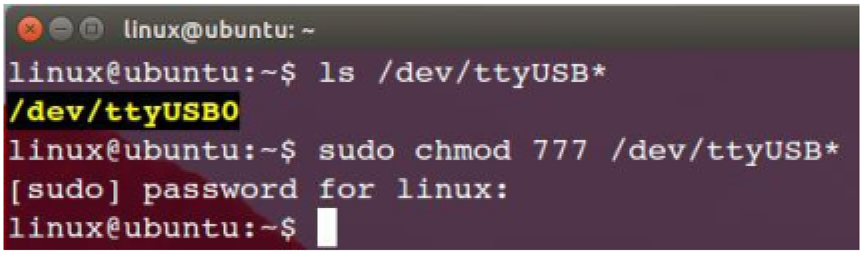
https://github.com/ldrobotSensorTeam/ldlidar_sl_sdk

2)git tool download operation

```
# First open the terminal interface, you can use the ctrl+alt+t shortcut
If you do not have git installed on your Ubuntu operating system, you can install
$ sudo apt-get install git

# Download the source code of the ROS2 feature pack:
$ cd ~
$ mkdir  ldlidar_ws
$ cd ~/ldlidar_ws
$ git clone https://gitee.com/ldrobotSensorTeam/ldlidar_sl_sdk.git
# or
$ git clone https://github.com/ldrobotSensorTeam/ldlidar_sl_sdk.git
```

6.5.2. Setting Device Permissions

First, the radar is connected to our switch module (CP2102 serial port switch module), and the module is connected to the computer. Then, open a terminal in ubuntu and type ls /dev/ttyUSB* to check whether the serial port device is connected. If a serial port device is detected, run the sudo chmod 777 /dev/ttyusb * command to grant the highest read/write permission to the file owner, group, and other users, as shown in the following figure.



6.5.3. Compile the source code

The source code is coded in C++11 and C language C99. The source code is compiled and constructed using tools such as CMake, GNU-make, and GCC. If you do not have the above tools installed on the Ubuntu system, run the following command to complete the installation.

```
$ sudo apt-get install build-essential cmake
```

If the preceding tools already exist in the system, perform the following operations.

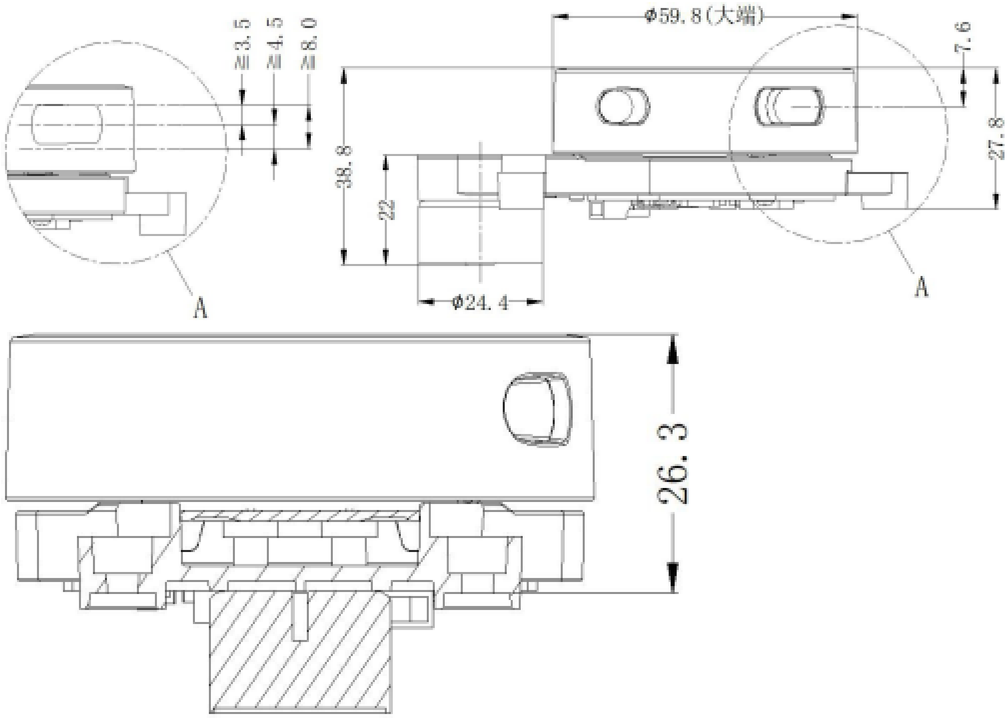
```
$ cd ~/ldlidar_ws/ldlidar_sl_sdk
..
$ mkdir build  # If the build folder does not exist in the ldlidar_sl_sdk directo
$ cd build
$ cmake ../
$ make
```

6.5.4. Run the program

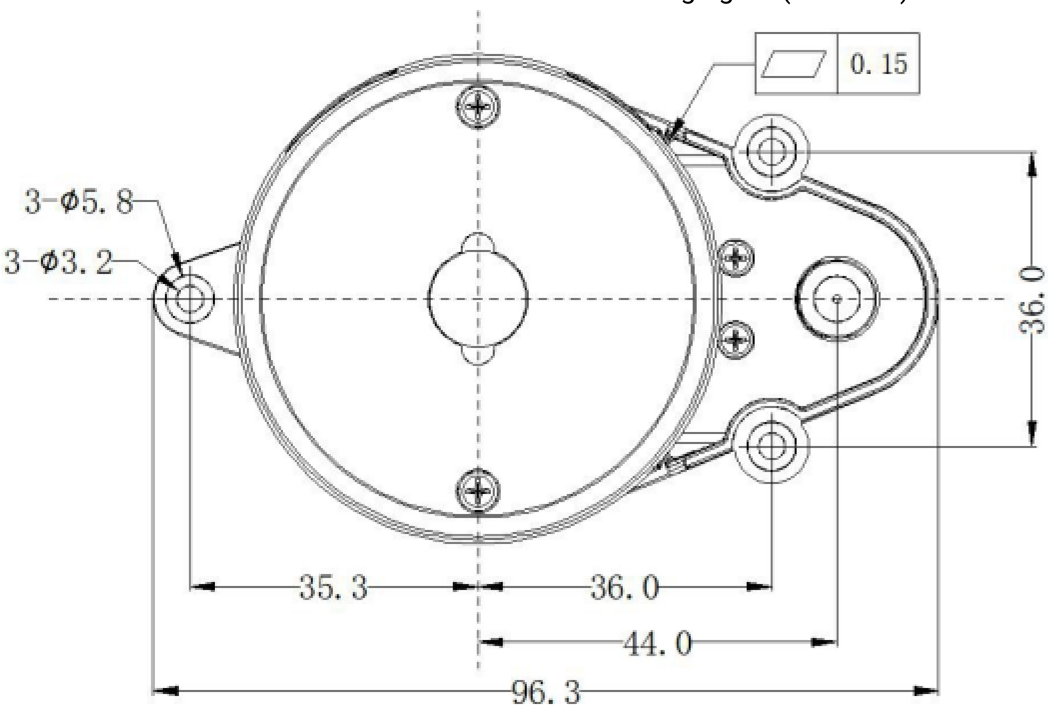
```
$ cd ~/ldlidar_ws/ldlidar_sl_sdk/build

$ ./ldlidar_sl  LD14  <serial_number>

#  such as ./ldlidar_sl  LD14  /dev/ttyUSB0
```

7. Optical window and mechanical dimensions

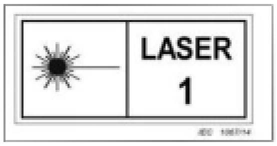
The laser transmitting and receiving in the ranging unit of LD14 requires an optical window, which needs to be exposed in the structure. The partial occlusion of the window by the external system will affect the ranging performance of LD14 to a certain extent. The following figure shows the optical window size (unit:mm). The overall dimensional tolerance is +0.3/-0.3mm.



Other installation dimensions are shown in the following figure (unit: mm) :



6. Safety and application scope



LD14 uses low-power infrared laser as the emission source, so it can ensure the safety of humans and pets. At present, this product has passed the Class I laser safety standard. LD14 complies with 21 CFR 1040.10 and 1040.11 except for deviations from Laser Circular No. 50 of 24 June 2007. Note: Adjusting or modifying this product may result in dangerous radiation exposure.

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